xMCF
ISO/PAS 8329

Description of mechanical connections and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
focus group on issues #105 / #61
2026-06-17

Meeting results are added in green font.

CONTACT Software

Carsten.Franke@contact-software.com
https://www.linkedin.com/in/diplmathdrcarstenfranke/

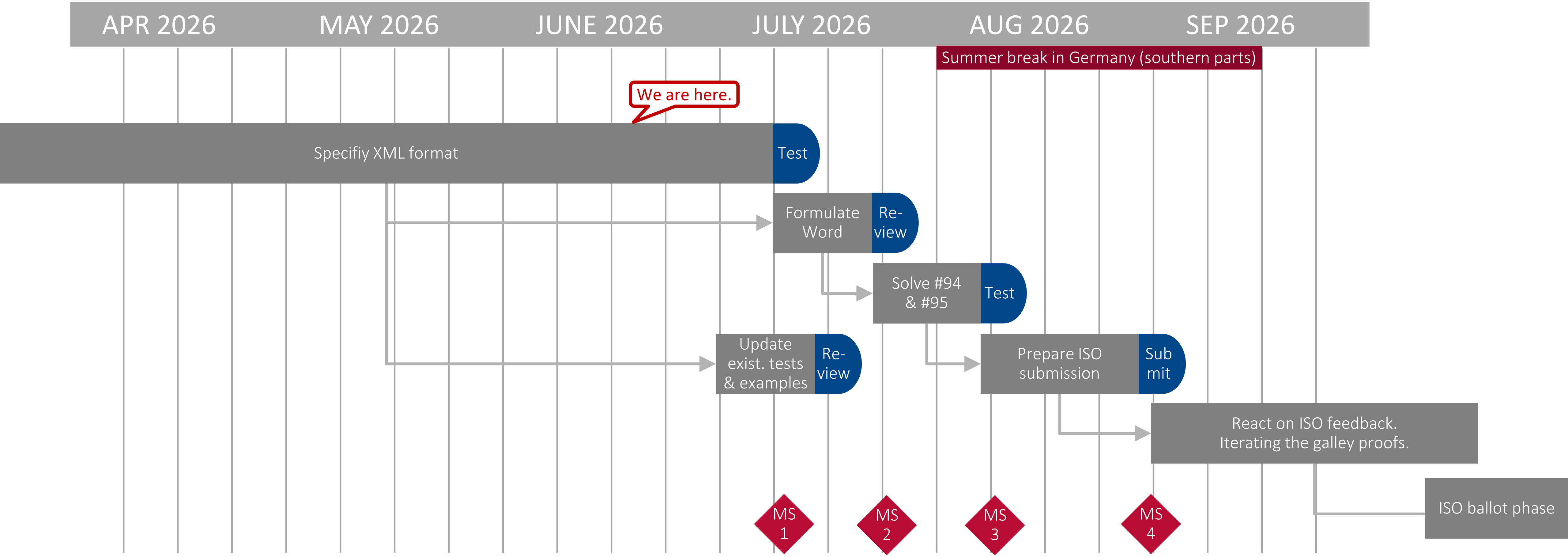
energizing great minds

Agenda

As announced at https://github.com/economidis-nick/createXSDforxMCF/blob/master/VDA_FAT_AK_25/Meetings/2026-06-17_VideoConference/README.md

- Welcome & Flying over today's topics
- Timeline for next release
- Work in progress
- Focus on XML / DOC suggestions, XSD- / Schematron-definition & examples
- Steps we should take next

Timeline for the next release



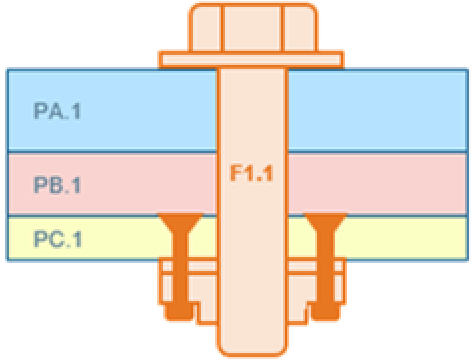
Updating the PAS (or else striving for an IS, directly) shall be cross-checked with the chair of ISO/TC 184 / SC 4.
Question was repeated at April 28th SC4/WG12 meeting. Though, no answer yet.

Issue #106: Support complex rivet-nut fasteners

as of 2026-06-12

Due to the composition of today's attendance list, the focus was placed on issue #106.

The task is to describe a design like this:



GH outlines a solution and invites discussion:

1. The nut is riveted to part PC.1.
   ⇒ One `connection_group` between PC.1 & nut.
2. The bolt connects PA.1, PB.1, and PC.1.
   The nut is modeled "`fixed_to`".

The discussion showed that this type of modelling is feasible. It can be described using the published standard.

GH will add a note to STEP's recommended practices and also post a comment on issue #106.
The issue will then be closed.

I created this slide *after* our meeting.
Is it correct, so far?
…

Conclusion of the Investigation

as of 2026-04-01

Combined XML verification by XML schema plus Schematron provides:

- Checks for several additional aspects, e.g. type confusion.
- Chance to separate business logic ($\mapsto$ Schematron) and formal syntax ($\mapsto$ Schema)

$\Rightarrow$ There are indisputable technical reasons to switch χMCF 3.2 validation to a Schematron-based method.

However:

- Implementing the complete Schematron file will take weeks.
- Many checks in XML schema need to be reduced from “mandatory” to “optional”. Will also take time.

NE & SH volunteer for both these items.

CF will care for updating the existing χMCF 3.2 example files. — Pending.

NE‘s recent commit and comment to issue #105

- Decision: robscan/gap is considered an attribute of the template.
- 2 FDS attributes pre_machined_hole_diameter & pilot_hole_diameter: both applicable
- Rivets: clearance, hole_diameter, hole_depth: applicable
- heat_stake/hole_diameter: applicable for template
- clip/hole_diameter & hole_length: applicable for template, since these are kind of “requirements” to the base part geometry, but not the actual feature sizes of that base part.
- rotational_joint/rotav/rotational_speed / compr. force: Habe been removed, before, see PDF.

- “Free text” confirmed. [Needs to be copied, to this slide set.]

Analysis: Which element attributes should (not) be allowed for the templates? (SH, continued from 2026-05-13)

- See file "Attributes_invalid_for_templates.pdf", as committed to GitHub 2026-05-19 21:41 UTC+2:00.
- Phrasing on slide 2 was reviewed. (bullet points 1–3)

<threaded_connection/>, <nut/> & <washer/> templates

as of 2026-04-29

Which sub-elements are allowed in a template? (NE)

Discussion of status of <threaded_connection/> templates.

- Make <nut/> and <washer/> appear in the <connection_templates/> section.
  Confirmed.

- Do we want <threaded_connection/> to appear in the <connection_templates/> section as well?
  Confirmed: yes.

- However, <connection_0d/> shall be *not* allowed in <connection_templates/> section.

- Should <rivet/> subtypes (<solid/>, <self-piercing/>, etc.)
  appear in the <connection_templates/> section?
  Decision: Only <rivet/> is allowed, not its sub-elements.

- Heat stakes, clips etc. are to be investigated later.

Directions (normal & tangential) in templates?

as of 2026-04-29

<u>Decision:</u>

- *No* directions shall appear inside a connection template.

<u>Rationale:</u>

- The combination of several connection templates with a number *N* of different directions delivers a high number (Cartesian product) of combined templates to be managed and hence cuts down the desired repetition count by the corresponding factor *N* (on average).
  Consequently, this approach is neither scalable nor cost-effective.
- Idea (investigation postponed): Introduce *separate* templates for certain directions, e.g. “**`negative_z`**”, *independent* from connection templates.

——————————————

- Also to be investigated later: Introduction of templates for `<custom_attributes/>`.

Recent results wrt. Schematron verification and `<threaded_connection>`.

as of 2026-05-13

See NE's commits of May 3, 2026:

- split the gumdrop_template.xml into 2 files
- schematron checks against <threaded_connection/>
- templates for rivet & bolt are introduced
  The following elements are introduced in <connection_templates/>:
  <rivet/>
  <washer/>
  <nut/>
  <threaded_connection/>
  Any attributes of these elements that were 'required', will now be checked by the schematron file, not by the xsd.
- Update README.md

Detailing XML Elements — Template Syntax, Screw Example

We identified the example XML file chapter_9_5_7_exampleA__patched_for_xMCF3.2.xml (provided by the comment of 2026-10-14 09:17+0100) as a good starting point.

If yes, the template must *not* have an attribute “label”, since this would usually be *unique to each single* connector instantiated from the template.

It suggests as a syntax for defining an (file-embedded) list of *connection templates* at XM[…]oot level:

For <washer/> and[…] we introduced the new attribute “[…] Shouldn’t we use it here, too? (I[…]

Decision, 2026-03-11:
We change to „template_label“ for all kind of templates.
➔ Draft example files to be updated.

```
<connection_templates>
    <connection_0d label="screw_template_4711">
        <threaded_connection part_code="M10x50 12.9" length="50." diameter="10"
            head_diameter="16.0" head_height="5" sink_size="4" thread_length="35" >
            <!-- Attributes "torque", "angle" and "pretension" must not appear in a template! -->
            <screw /> <!-- Screw may come without any attributes -->
            <washer outer_diameter="20"/>
        </threaded_connection>
    </connection_0d>
</connection_templates>
```

Nick and Stephan’s discussion at issue #105 may shed a new light on attribute naming.
→ See according slide!

pending

Homework: Consider what “misuse” or contradictory set of data can occur by this approach!

Detailing XML Elements — Reference Syntax, Screw Example

The example suggests as syntax for *referring to* a connection template at <connection_0d/> level:

Default form — no change to / overwriting template data:

```
<!-- Screw exactly as defined in connection_templates: -->
<connection_0d label="SCREW_100532" template="screw_template_4711">
    <threaded_connection>
        <normal_direction x="2.9" y="0.1" z="0.0" />
    </threaded_connection>
    <loc> 1500.3809 838.75885 730.6529 </loc>
</connection_0d>
```

Minimal change — overwriting an attribute value:

```
<!-- Screw exactly as defined in connection_templates, but with smaller torque: -->
<connection_0d label="SCREW_100533" template="screw_template_4711">
    <threaded_connection torque="70" >
        <normal_direction x="3.0" y="0.0" z="0.0" />
    </threaded_connection>
    <loc> 1540.3809 838.75885 730.6529 </loc>
</connection_0d>
```

pending

Detailing XML Elements — Reference Syntax, Screw Example

The example suggests as syntax for *referring to* a connection template at `<connection_0d/>` level:

<u>Overwriting a child element:</u>

```
<!-- Screw exactly as defined in connection_templates, but with larger washer: -->
<connection_0d label="SCREW_101537" template="screw_template_4711">
    <threaded_connection>
        <normal_direction x="3.1" y="-0.1" z="0.0" />
        <washer outer_diameter="30"/>
    </threaded_connection>
    <loc> 1580.3809 838.75885 730.6529 </loc>
</connection_0d>
```

Attention: Here, only the `outer_diameter` of the `<washer/>` is overwritten. Any other washer-parameter given from the template stays effective.

pending

Detailing XML Elements — Template Syntax, Bolt Example

Nick and Stephan's discussion at issue #105 may shed a new light on attribute naming.
→ See according slide!

pending

The example extends to cases of *deeper structured* elements:

For <washer/> and <nut/>,
we introduced the new attribute "template_label".
Shouldn't we use it here, too? (Instead of "label"). (Cf. slide 7.)

```
<connection_templates>
    <connection_0d label="bolt_template_4712">
        <threaded_connection part_code="M10x50 12.9" length="50" diameter="10"
            head_diameter="16.0" head_height="5" sink_size="4" thread_length="35" >
            <!-- Attributes "torque", "angle" and "pretension" must not appear in a template! -->
            <normal_direction x="0" y="0" z="-10"/>
            <washer outer_diameter="20" inner_diameter="10.3" thickness="1.5" attached="true"/>
            <!-- Washer attached to the screw head -->
            <bolt>
                <nut diameter="16." height="5" />
                <!-- Friction attributes must not appear in a template,
                     unless it is against an attached washer -->
                <!-- Attributes "clipped_to" or "fixed_to" must not appear in a template,
                     since they each contain a flange partner index! -->
            </bolt>
        </threaded_connection>
    </connection_0d>
</connection_templates>
```

Detailing XML Elements — Reference Syntax, Bolt Example

The connection template can be *referred to* for example like this:

```
<!-- Bolt exactly as defined in connection_templates, but with less torque, larger washer & nut: -->
<connection_0d label="BOLT_100732" template="bolt_template_4712">
    <threaded_connection torque="70">
        <normal_direction x="3.1" y="-0.1" z="0.0"/>
        <washer outer_diameter="30"/>
```

Check, whether a washer may refer to a connection template, too! → we need examples to better understand. → next slides!

```
        <bolt>
            <nut height="8"/>
```

Check, whether a nut may refer to a connection template, too! → we need examples to better understand. → next slides!

```
        </bolt>
    </threaded_connection>
</connection_0d>
```

pending

Rule (→ Word file): If the connection template is a <screw/>, the referring element must not redefine it to a <bolt/> and vice versa!
To be extended to similar situations in case of future extensions of χMCF, e.g. a spotweld (from template) must not be redefined to a gumdrop!

Detailing XML Elements — Template Syntax for a Washer

Draft example:

```
<!-- The new parameter "template_label" is allowed for a template, only. -->
<washer template_label="template for a loose washer"
    outer_diameter="20" inner_diameter="10.3" thickness="1.5"
    strength_property_class="12.9" part_code="M10x20x1.5 12.9" />
    <!-- Washer next to nut would have friction to last part, which is unknown. Thus, not allowed. -->
    <!-- A washer can be attached to a screw head or a nut.
         But here as a template without context, one can't know. Thus, "attached" is not allowed. -->
```

Remarks:

- Templates must not contain friction parameters, because friction always is a physical property of a *combination* of two items.
- We should differentiate between “firmly fixed” washers (i.e. washer and nut/screw form one solid)
  and “rotate-able” ones, where the washer and nut/screw can rotate independently around their common axis.
  Remark: BETA CAE’s study (2025-02-18) showed that we do *not* need a new attribute, here.

pending

Detailing XML Elements — Template Syntax for a Nut w/ Washer

Draft example:

```
<!-- The new parameter "template_label" is allowed for a template, only. -->
<nut template_label="template for a nut with an attached washer"
    diameter="16." height="5" strength_property_class="12.9" part_code="M10x5 nut 12.9"
    static_friction="0.8" kinetic_friction="0.6" >
    <!-- Attributes "torque" and "angle" must not appear in a template! -->
    <!-- Attributes "clipped_to" or "fixed_to" must not appear in a template,
         since they each contain a flange partner index! -->
    <!-- The washer of this nut is attached to it in such a way that it can rotate.
         Therefore, the nut's friction against the washer is meaningful and allowed. -->

    <!-- If the washer is part of or firmly attached to the nut, it cannot have a part code! -->
    <washer outer_diameter="25" thickness="1.5" strength_property_class="12.9"
        attached="true" static_friction="0.7" kinetic_friction="0.5"/>
    <!-- The friction of the washer describes the contact to the connected part. -->
    <!-- This washer could reference a template, itself. -->
</nut>
```

Remark: Friction between nut and attached but rotate-able washer is allowed!

pending

Detailing XML Elements — Template Syntax for a Bolt, using Washer & Nut templates, itself.

For <washer/> and <nut/>,
we introduced the new attribute “template_label”.
Shouldn’t we use it here, too? (Instead of “label”). (Cf. slide 7.)

Nick and Stephan’s discussion at issue #105
may shed a new light on attribute naming.
→ See according slide!

pending

Draft example:

```
<!-- The next bolt t… …te references washer and nut templates, itself: -->
<connection_0d label="bolt_template_4733">
    <threaded_connection length="50" diameter="10" head_diameter="16" head_height="5" thread_length="35"
                         part_code="M10x50 12.9">
        <!-- Attributes "torque", "angle" and "pretension" must not appear in a template! -->
        <normal_direction x="0" y="0.8" z="-1.2"/>
        <washer template="template for a loose washer"/>  <!-- The washer next to the bolt's head. -->
        <bolt>
            <nut template="template for a nut with an attached washer"/><!-- Nut has attached washer.-->
        </bolt>
    </threaded_connection>
</connection_0d>
```

Detailing XML Elements — Template Syntax, Gumdrop Example

(new — see file “chapter_10_5_exampleB__with_templates.xml”)

For a gumdrop, the example template would be:

```
<connection_templates>
    <connection_0d label="GumDrop template">
        <!-- Assumed Unit system with mass attribute with value="kg" -->
        <gumdrop diameter="5.0" mass="0.0033" material="Glue_Material" />
    </connection_0d>
</connection_templates>
```

For <washer/> and <nut/>,
we introduced the new attribute “template_label”.
Shouldn’t we use it here, too? (Instead of “label”). (Cf. slide 7.)

Nick and Stephan’s discussion at issue #105
may shed a new light on attribute naming.
→ See according slide!

pending

Contentual Action Items ≙ Next Steps:

- **<u>Standard document (Word file):</u>**
  - Description of overall concept & objectives, intended application & motivation etc.
  - Description of semantics for internal catalog
  - Description of semantics for reference to a connection template
  - Definition of example use cases and in-document XML-examples
- **<u>XML & XSD</u>** — See discussion in [#105](#105)!
  - Sketching XML example file(s), incl. (w.i.p.)
- precedence rule (w.i.p.) and dropped mandatory attribute,
- internal (w.i.p.) and external file-based catalog
  - Description of syntax for internal catalogs (w.i.p.)
  - Description of syntax for reference to a connection template:
- internal (w.i.p.), external file-based, external database-based
  - Drafting XML schema (XSD) and test cases / test example files
Homework: Collect use cases which can yield e.g. test cases, shared as Markdown files in GitHub at https://github.com/economidis-nick/createXSDforxMCF/tree/issues_61_94_95_105/VDA_FAT_AK_25/Meetings/2026-02-18_VideoConference pending
  - Investigate χMCF elements and attributes for eligibility for template.
⇒ we need a rule, here! ⇒ Homework! (cf. Tables 35—79 of the standard) pending

Next meetings

(Suggestions in light blue.)

- 2026-06-24, 16:30 CEST — focus group on #105 / #61
  — objective: agree on XML suggestions & examples

- 2026-07-01, 15:00 CEST — focus group on #105 / #61
  — objective: agree on XML suggestions & examples

- 2026-07-08, 15:00 CEST — focus group on #105 / #61
  — objective: t.b.d.

- 2026-07-15, 15:00 CEST — focus group on #105 / #61
  — objective: t.b.d.

Suggested time slots have been confirmed.

CF sent / will send out invitations.